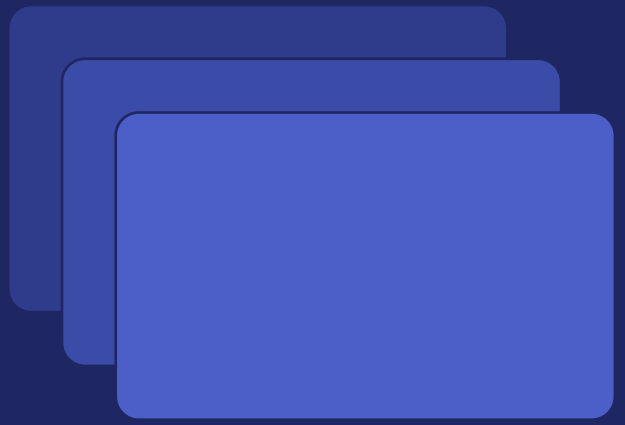

Container and Image Security in Docker

Session Overview; What We will Build, Break, and Harden

What We'll Cover

1. **Image Security**
   Hygiene, digest pinning, vulnerability scanning
2. **Docker Hardened Images**
   Minimal, low-CVE drop-in replacements
3. **Secrets Management**
   Keeping credentials out of image layers
4. **Multi-Stage Builds**
   Separating build tools from runtime
5. **Dockerfile Best Practices**
   Non-root, read-only, pinned versions
6. **Runtime & Provenance**
   Capabilities, SBOMs, no privileged mode

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یک مرور کوچول

یک مرور کوچول

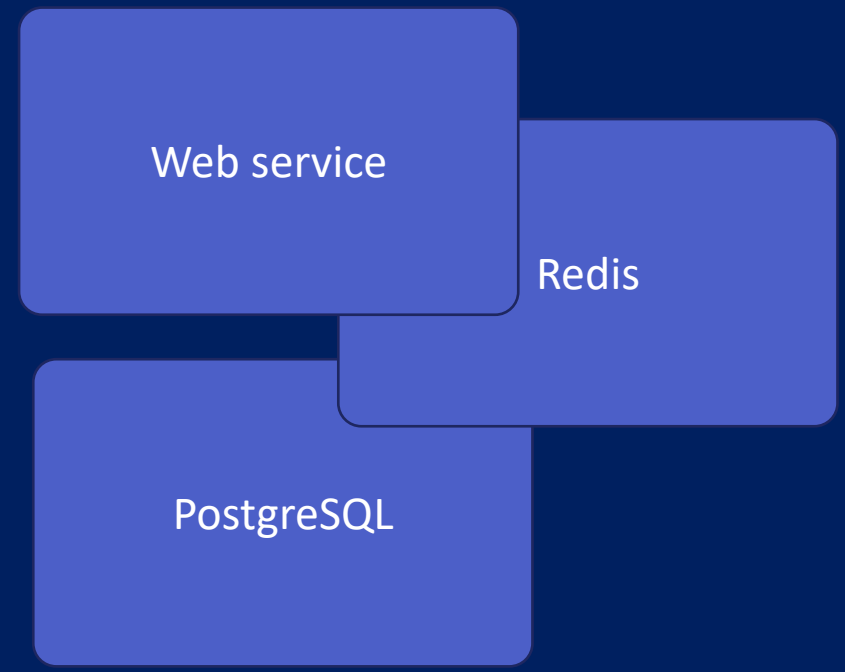
Dockerfile

```
docker build -t fozouni/compose-demo:v1 .
```

Docker Image

Docker compose

```
docker compose up -d
```



ممکن است نیاز به مرور چندین باره‌ی مطالب این فایل پیدا کنید.

موارد مطرح شده در ادامه تنها جهت ارائه‌ی یک کلیت از کارهای پیش رو هستند.

1- Image Security

Periodic Re-Scans

Base Image Hygiene

Vulnerability Scanning

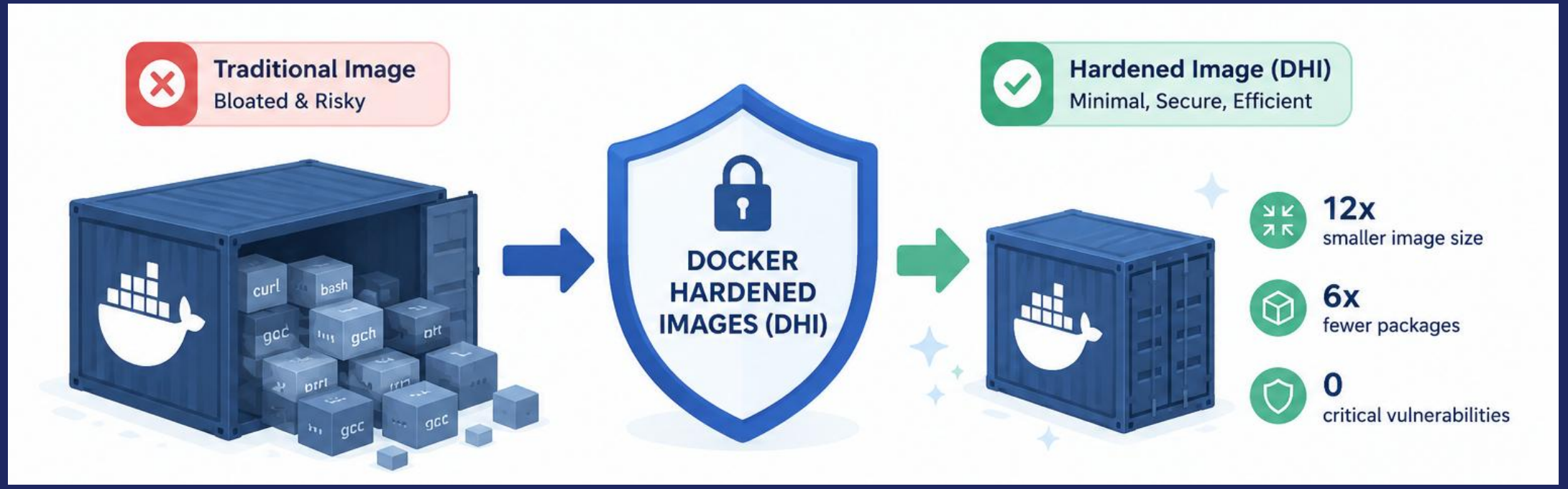
Rebuild Regularly

1- Image Security

Start clean, stay patched, catch problems before they ship

- **Base Image Hygiene** — Minimal official images, pinned by digest; tags are mutable, digests aren't.
- **Rebuild Regularly** — A "frozen" image silently rots as new CVEs surface upstream. Rebuild to repull patches.
- **Vulnerability Scanning** — Trivy & Docker Scout scan OS packages + language deps; fail CI on Critical/High findings.
- **Periodic Re-Scans** — New CVEs are disclosed after the fact; re-scan images already sitting in the registry.

2- Docker Hardened Images (DHI)



2- Docker Hardened Images (DHI)

Minimal, hardened, drop-in replacements

12x

smaller image size

6x

fewer packages

0

critical vulnerabilities

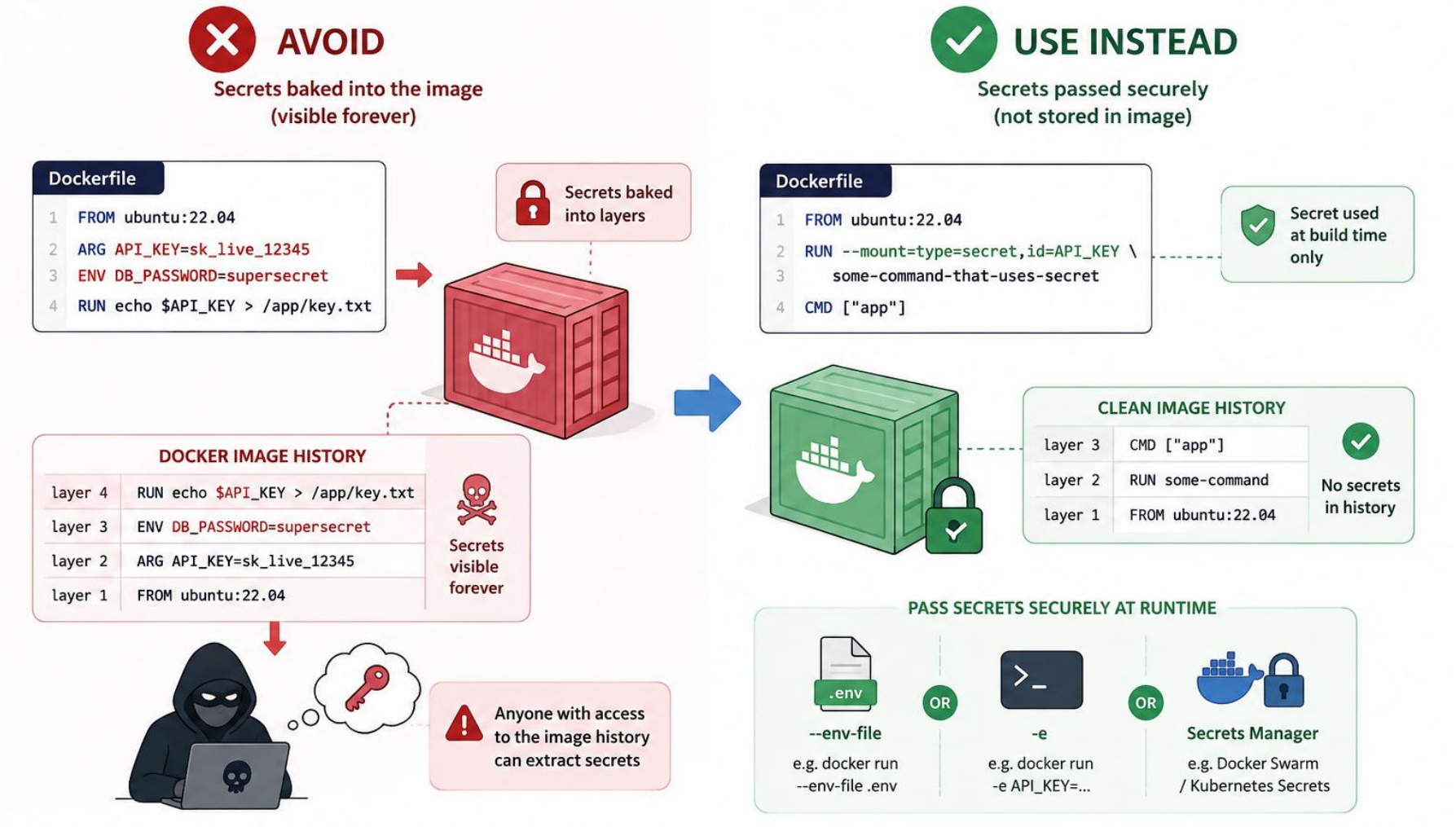
Two variants for every language image:

dev --- build tools, shell, root user (build stage)
runtime --- stripped down, production-ready

چرا این دو تا جدا شدن؟

برای اینکه در مرحله‌ی ساخت (build) به ابزار زیاد نیاز داریم، اما در محیط اجرای نهایی (runtime) فقط به حداقل فایل‌ها نیاز داریم تا حمله‌پذیری کمتر، حجم کمتر و سرعت بالاتر برود .این یعنی اصل حداقل دسترسی (Principle of Least Privilege) در امنیت.

3- Secrets Management



3- Secrets Management

Once a secret is in a layer, it's in the image history forever

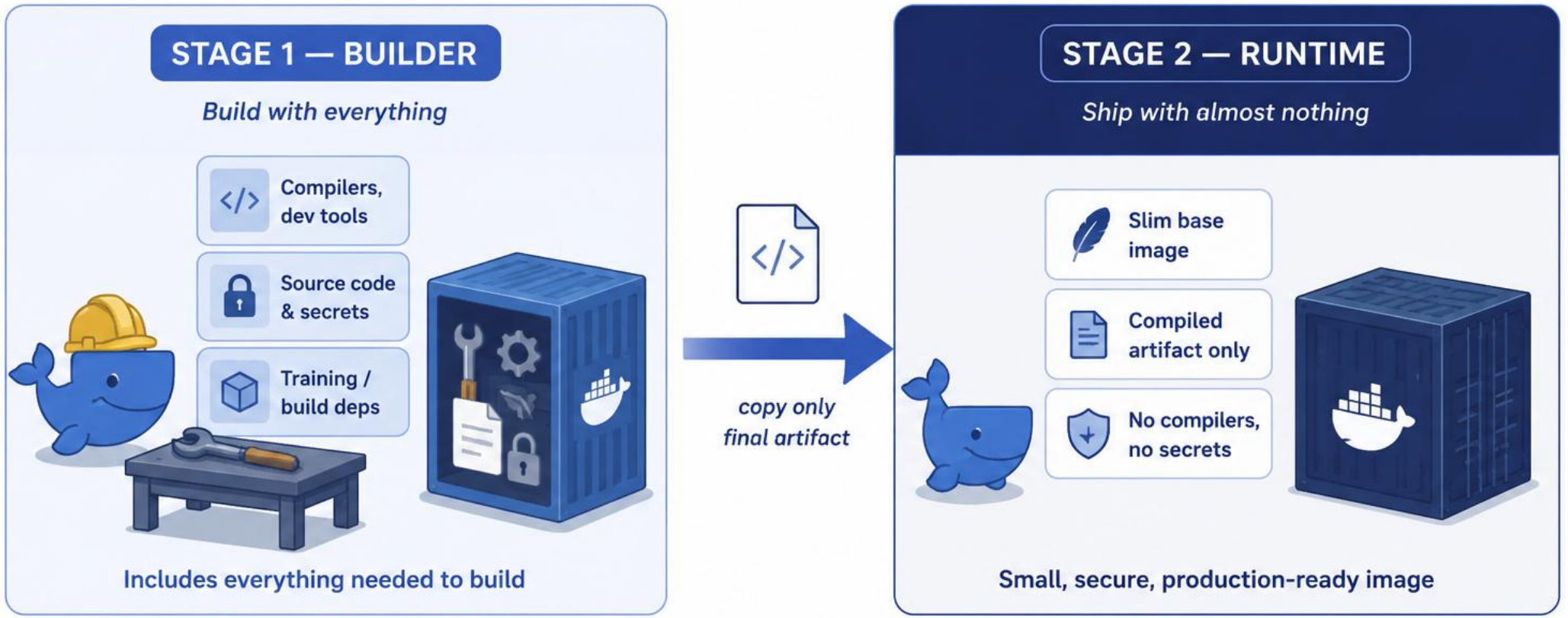
✕ Avoid

- COPY .env / keys / tokens into an image
- ARG / ENV for credentials --- visible via docker history
- Assuming RUN rm secret.txt removes it (it doesn't)

✓ Use Instead

- BuildKit --mount=type=secret at build time
- Pass tokens via --env-file or -e at runtime
- .dockerignore for .env, keys, and datasets

4- Multi-Stage Builds

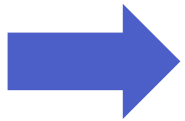


4- Multi-Stage Builds

Build with everything, ship with almost nothing

Stage 1 --- Builder

Compilers, dev tools
Source code & secrets
Training / build deps



copy only final artifact

Stage 2 --- Runtime

Slim base image
Compiled artifact only
No compilers, no secrets

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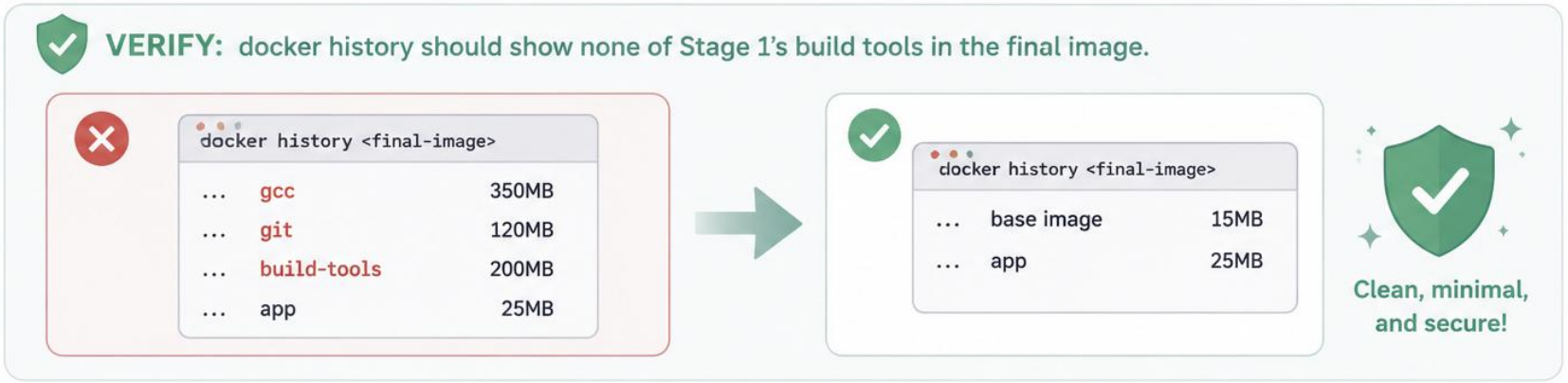

5- Dockerfile Best Practices

1 Non-Root User

USER appuser; never leave containers running as root.

2 Read-Only Filesystem

--read-only at runtime, with explicit writable volumes for /tmp, caches.

3 .dockerignore

Exclude .git, .env, datasets, notebooks — keep secrets out of the build context.

4 No Remote ADD

Prefer COPY + checksum-verified downloads over ADD with remote URLs.

5 Pin Versions

pip install torch==2.3.1; avoid floating versions and dependency confusion.

6 Verify Checksums

Validate signatures/hashes of downloaded models or datasets at build time.

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6- Runtime Security & Provenance

Runtime Hardening

--cap-drop=ALL

Drop all Linux capabilities,
add back only what's needed

--security-opt=no-new-privileges

Blocks setuid escalation (sudo, su)

Never --privileged

Disables container isolation entirely

Supply-Chain Provenance

SBOM (Software Bill of Materials)

Tracks every dependency shipped
in the image

docker scout / trivy --format cyclonedx

Generate & store as a build artifact
for audits

Docker socket

Never mount it into a container –
it's root on the host

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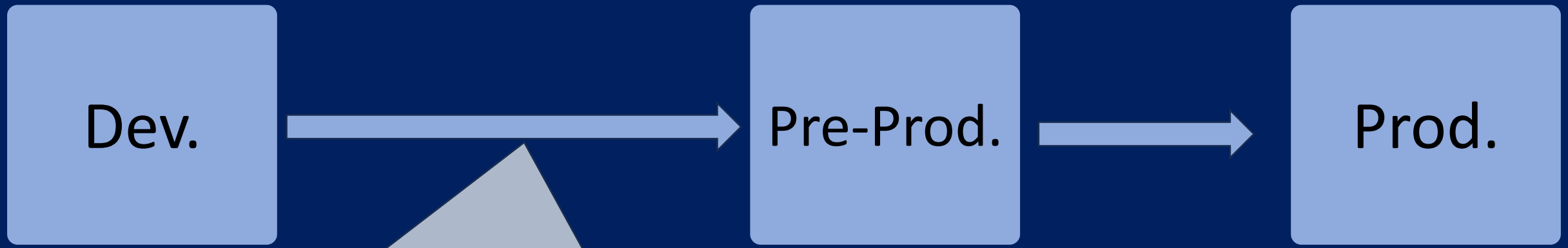
اس‌بی‌اوام به شما می‌گوید «چه چیزی، با چه نسخه‌ای، در کجاست» تا وقتی خطری پیش می‌آید، بیدرنگ بدانید آیا تحت تأثیر قرار گرفته‌اید یا نه و چطور رفعش کنید. بدون این مولفه انگار یک جعبه‌ی سیاه داریم که نمی‌دانیم داخلش چیست.

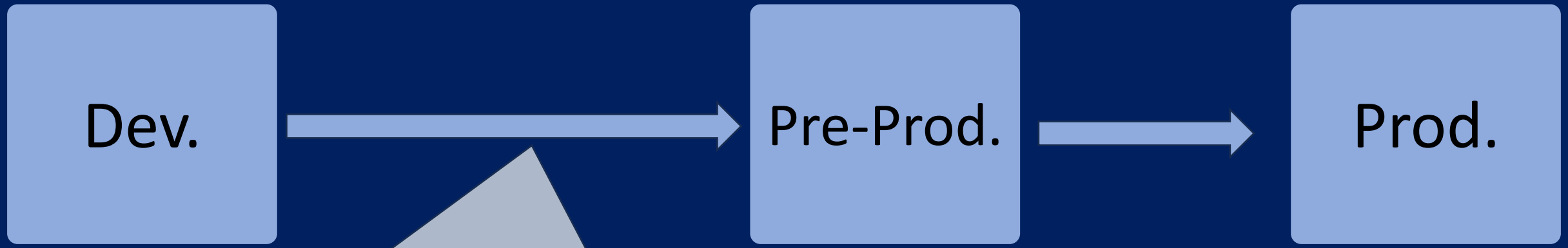
Whole picture of this story

Whole picture of this story



در این مرحله هست که باید سعی کنیم همه چیز رو اتومیت یا خودکار کنیم. از جمله:

- بررسی خودکار ایمیج‌ها
- وجود رمز یا سیکرت خاص داخل آنها یا سورس کد قبل از ارسال به گیت‌هاب
- چک کردن مسائل امنیتی اپ پایتونی
- اسکن فایل و دایرکتوری‌ها
- ...



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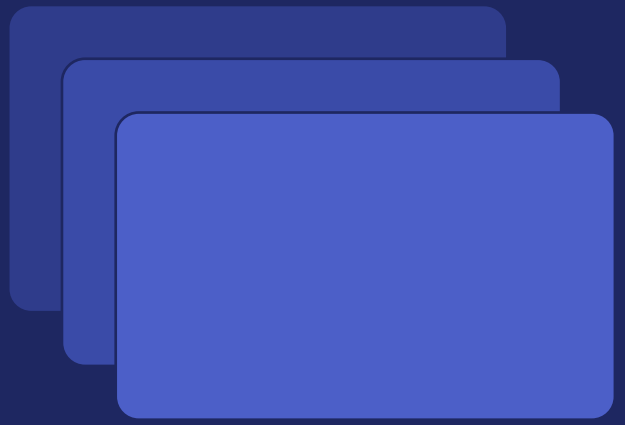
- اسکن فایل و دایرکتوری‌ها
- ...

این خودکار سازی چک نمودن ابزارهای موجود در سرویس از DevSecOps میاد. ما در MLSecOps دقیقن همین کارها رو باید برای مدل‌ها و دیتای موجود در سیستم خودمون هم تنظیم کنیم که انجام شوند. وگرنه یه جای سیستم خواهد لنگید.

تمرکز کنید روی کاری که انجام می‌شود.
نه ابزار خاص انجام آن کار.

کلیت کار هیچگاه تغییر نخواهد کرد. اما ابزارها، بله.

Now let’s goooo for practical stuffs in the “.md” file 🚀



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